Problems You Need to Solve with Your Database Design

1. Lack of Unique Identifiers

Currently, we have no guaranteed unique IDs for patients, doctors, departments, or appointments.

Introduce something that ensures uniqueness.

2. Disconnected Relationships

In Excel, appointments are listed, but there is no enforceable link to valid patients or doctors.

Implement measures to maintain referential integrity between patients, doctors, departments, and appointments.

3. Invalid or Ambiguous Data Entries

For example, gender values like "X", appointment statuses like "On Hold", and inconsistent date formats.

Allowed values (Gender must be 'M', 'F', or 'O'; Status must be 'Scheduled', 'Completed', or 'Cancelled').

4. Unregulated Scheduling

Doctors are occasionally double-booked, and appointments are being scheduled in the past.

Design and implement measures to automatically prevent invalid appointment entries during insertion.

5. Open Access to Sensitive Patient Information

All doctors currently see all data, regardless of their role or department.

Create limitations that allow access to data based on credentials and roles provided by the firm. Only senior doctors can view all patients in their department, else doctors can only see details (medication, appointment, reports) of their respective patients.

6. Disconnected Reporting

- There's no way to generate billing or departmental summaries across the hospital.
- Implement a way that generates **monthly revenue reports by department**.